

Unified Priority Statement: Quartz Birefringence for Controlled Mass Splitting and Vacuum Excitation – Validated by Toronto Superconductor (2024) and Truncated Photon Effect (2026)

Author: Peter James Thompson

ORCID: 0009-0005-8688-7507

Date: 4 June 2026

Contact: peterjamesthompson101@gmail.com

Abstract

This paper consolidates three independent lines of evidence that validate the core principle of my 2018 manuscript “Physics of light, time and dimensions”: splitting a light wave via quartz birefringence creates two independent pathways that can (1) separate a mass from its paired potential ($m + m' = 0$), enabling high-temperature superconductivity, and (2) excite quantum vacuum fluctuations to generate infinite photon swarms. The 2024 Nature Materials paper by the University of Toronto team (Ni-Bi van der Waals heterostructure) experimentally demonstrates the mass-splitting mechanism in layered interfaces – the same principle I described using quartz and copper-oxide nanoparticles. The 2026 Physical Review Letters paper by Rukan et al. (truncated photon effect) theoretically shows that abruptly splitting a photon’s wave packet creates an infinite photon swarm – a phenomenon my birefringent quartz design can implement continuously and controllably. My 2018 work predates both by six to eight years and provides a practical engineering platform. This statement establishes priority and invites acknowledgment.

1. The 2018 Birefringent Quartz Design

In my 2018 manuscript “Physics of light, time and dimensions” (pages 20-27), I described:

- A quartz crystal (birefringent material) illuminated by an infrared laser.
- The birefringence splits the light wave into two independent, orthogonally polarised waves (ordinary and extraordinary rays) that propagate on a double-helix path.
- Copper-oxide nanoparticles embedded in the quartz create an internal potential barrier, but the splitting itself is purely optical.
- This splitting enables two distinct physical effects:
 1. Mass splitting for high-temperature superconductivity: the two pathways separate the forward-time mass component from its reverse-time paired potential ($m + m' = 0$), eliminating resistance.
 2. Vacuum excitation: the two waves interact with quantum vacuum fluctuations, generating additional photons.

The manuscript therefore contains the core principle of using wave-packet splitting to control both matter and vacuum states.

2. Validation 1: Toronto Superconductor (Nature Materials, 2024)

In 2024, a University of Toronto team published a paper in Nature Materials on a Ni-Bi van der Waals heterostructure – a layered material with a sharp interface that exhibits high-temperature

superconductivity. The key mechanism is mass splitting: the interface separates the charge carriers into two correlated populations, effectively creating $m + m' = 0$. This is exactly the principle I described in my 2018 design, but realised with chemical layers instead of an optically split wave.

Significance: The Toronto paper provides independent, peer-reviewed experimental validation that layered interfaces can produce high-temperature superconductivity via mass splitting. My 2018 manuscript predates it by six years and describes the same mechanism using a different physical implementation (quartz birefringence). Therefore, I claim priority for the underlying principle.

Clarification on copper oxide: In my design, copper-oxide nanoparticles are used to create a potential barrier and to enable charge storage (for capacitors/batteries). They are not required for the superconducting effect itself. Any sharp interface – optical (birefringent split) or material (Ni-Bi layers) – can produce mass splitting.

3. Validation 2: Truncated Photon Effect (PRL, 2026)

In 2026, Rukan, Skaar and colleagues published a theoretical study in Physical Review Letters showing that abruptly truncating a single photon's wave packet (e.g., with an ultra-fast shutter) excites quantum vacuum fluctuations and creates an infinite superposition of photons (an "infinite swarm"). The effect relies on splitting the wave packet into transmitted and blocked parts.

My 2018 birefringent quartz design achieves the same splitting continuously and controllably, without an abrupt shutter. By adjusting crystal thickness, orientation, or temperature, one can tune the degree of splitting and thus the number of photons generated from the vacuum. Moreover, the two output waves can be recombined or manipulated, allowing engineering of the quantum state.

Significance: The PRL paper provides theoretical validation that splitting a light wave can harvest vacuum fluctuations to produce many photons. My design not only anticipates this effect but also offers a practical, tunable device to realise it. My 2018 work predates the PRL paper by eight years.

4. Synthesis: A Unified Principle

Both the Toronto superconductor and the PRL truncated photon effect derive from the same foundational concept: splitting a coherent wave (or wave packet) into two separate components creates a situation where the original system's identity is distributed, allowing new emergent phenomena – mass cancellation in superconductivity and photon generation from the vacuum. My 2018 birefringent quartz design is the first written description of this unified principle, and it provides a practical means to achieve both effects controllably.

Effect Toronto 2024 PRL 2026 My 2018 work

Mass splitting ($m + m' = 0$) Experimental (Ni-Bi layers) Not addressed Described (quartz + CuO)
Vacuum excitation / infinite photon swarm Not addressed Theoretical (abrupt shutter) Described (birefringent split)

Practical, tunable device No No Yes (quartz, IR laser, CuO)

Priority date 2024 2026 2018

5. Conclusion

My 2018 manuscript “Physics of light, time and dimensions” contains the first written description of using wave-packet splitting (via quartz birefringence) to achieve both high-temperature superconductivity (mass splitting) and controlled vacuum excitation (photon swarm generation). Subsequent independent work – the Toronto 2024 Nature Materials paper and the 2026 PRL truncated photon paper – experimentally and theoretically validate the same principles, without citing my prior art. I therefore assert priority for these concepts and request that the scientific record be corrected to acknowledge my 2018 work.

References

1. Thompson, P.J. (2018). Physics of light, time and dimensions - Pages 20-27. Timestamped PDF available from the author.
2. University of Toronto team (2024). “Ni-Bi van der Waals heterostructure for high-temperature superconductivity”. Nature Materials. DOI: (to be added).
3. Rukan, I.C.O., Skaar, J., et al. (2026). “Truncated Photon”. Physical Review Letters. DOI: (to be added). News summary: Jarman, S. (2026). “Cutting a photon in two creates an infinite swarm of particles”.
4. Thompson, P.J. (2025). The Harmonic Framework of Reality Rev III. ISBN 978-1-300-04142-9.

Correspondence: peterjamesthompson101@gmail.com

This document is a public Proity Record.